elemental copper is more stable than Cu^{2+}. There are two deviations., In the general trend, Fig shows that $\left(E^0_{M^{2+}/M}\right)$ value for manganese and zinc are more negative than the regular trend. It is due to extra stability which arises due to the half filled d^5 configuration in Mn^{2+} and completely filled d^{10} configuration in Zn^{2+}.

Transition metals in their high oxidation states tend to be oxidizing . For example, Fe^{3+} is moderately a strong oxidant, and it oxidises copper to Cu^{2+} ions. The feasibility of the reaction is predicted from the following standard electrode potential values.

$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+} \quad E^0 = 0.77V$

$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s) \quad E^0 = +0.34\ V$

The standard electrode potential for the M^{3+}/M^{2+} half-cell gives the relative stability between M^{3+} and M^{2+}. The reduction potential values are tabulated as below.

Reaction	Standard reduction potential (V)
$Ti^{3+} + e^- \longrightarrow Ti^{2+}$	–0.37
$V^{3+} + e^- \longrightarrow V^{2+}$	–0.26
$Cr^{3+} + e^- \longrightarrow Cr^{2+}$	–0.41
$Mn^{3+} + e^- \longrightarrow Mn^{2+}$	+1.51
$Fe^{3+} + e^- \longrightarrow Fe^{2+}$	+0.77
$Co^{3+} + e^- \longrightarrow Co^{2+}$	+1.81

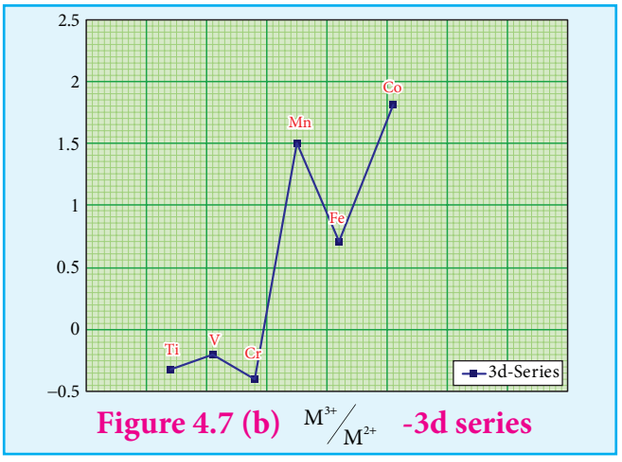


Figure 4.7 (b) M^{3+}/M^{2+} **-3d series**

The negative values for titanium, vanadium and chromium indicate that the higher oxidation state is preferred. If we want to reduce such a stable Cr^{3+} ion, strong reducing agent which has high negative value for reduction potential like metallic zinc $\left(E^0 = -0.76\ V\right)$ is required.

The high reduction potential of Mn^{3+}/Mn^{2+} indicates Mn^{2+} is more stable than Mn^{3+}. For Fe^{3+}/Fe^{2+} the reduction potential is 0.77V, and this low value indicates that both Fe^{3+} and Fe^{2+} can exist under normal conditions. The drop from Mn to Fe is due to the electronic structure of the ions concerned.Mn^{3+} has a $3d^4$ configuration while that of Mn^{2+} is $3d^5$. The extra stability associated with a half filled d sub shell makes the reduction of Mn^{3+} very feasible ($E^0 = +1.51V$).

4.3.6 Magnetic properties

Most of the compounds of transition elements are paramagnetic. Magnetic properties are related to the electronic configuration of atoms. We have already learnt in XI STD that

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